

SOCIOL 723: Social Statistics II

Week 5, Potential Outcomes and DAGs

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Today

The Potential Outcomes Framework

From Path Analysis to DAGs

Reading a DAG

When Conditioning Goes Wrong

Good Controls and Bad Controls

The Front-Door Criterion

Defining a Well-Posed Estimand

★ = essential, you must understand this

★ = for understanding, not examined

From Estimation to Causal Questions

Weeks 1–4 were about *estimation*: given a target parameter, how do we estimate it and how uncertain are we? OLS estimates the best linear approximation to a conditional expectation; maximum likelihood estimates the parameters of a probability model. Neither says the word “effect.”

Today's question

What are we trying to estimate, and what would have to be true about the world for our estimate to recover it?

Two languages answer it, and they are complementary:

- **Potential outcomes**: what would have happened to each unit under each treatment. Precise about the estimand.
- **Directed acyclic graphs**: how the variables cause one another. Precise about what to condition on.

Everything in Weeks 6–13 applies one or both.

Identification Is Not Estimation

Manski (1995)

“Studies of identification seek to characterize the conclusions that could be drawn if one could obtain an unlimited number of observations. Identification problems cannot be solved by gathering more of the same kind of data.”

- **Estimation** (Weeks 1–4): uncertainty from a finite sample; it shrinks as n grows.
- **Identification** (today): whether the quantity you want is pinned down by the distribution of what you observe, even with $n = \infty$. If not, only an assumption resolves the ambiguity.
- **Identification analysis**: state assumptions about how the data were generated and show they carry you from observed to potential outcomes. An **identification strategy** is a labelled source of variation in the treatment plus the technique that exploits it (Angrist and Krueger 1999).

The rest of the course is identification analysis, design by design.

Potential Outcomes

Two Worlds Per Person ★

For a binary treatment $D_i \in \{0, 1\}$, each unit i has *two* outcomes written into it before treatment is assigned:

$Y_i(1)$ = the outcome if unit i were treated,

$Y_i(0)$ = the outcome if unit i were not treated.

The **individual treatment effect** is $\tau_i = Y_i(1) - Y_i(0)$. We observe only one of the two:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) \quad (\text{the switching equation})$$

Holland's fundamental problem of causal inference (Holland 1986)

τ_i is never observed for anyone. Causal inference is a **missing data problem**, and every method in this course is a strategy for filling in the missing column.

The Missing Data View

Unit	D_i	$Y_i(0)$	$Y_i(1)$	τ_i	Y_i (observed)
1	1	?	12	?	12
2	0	7	?	?	7
3	1	?	9	?	9
4	0	5	?	?	5

- Half the table is missing *by construction*, not by bad luck. No larger sample fixes it.
- So we give up on τ_i and target *averages*, which are estimable under assumptions.
- “No causation without manipulation” (Holland 1986): if you cannot even imagine intervening to set D_i , the potential outcomes are not well defined. “The effect of race” or “of gender” needs restating as the effect of a manipulable treatment, such as the race signalled on a job application (VanderWeele and Robinson 2014).

Estimands: Say Which Average You Want ★

$ATE = E[Y_i(1) - Y_i(0)]$ average treatment effect

$ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1]$ on the treated

$ATU = E[Y_i(1) - Y_i(0) \mid D_i = 0]$ on the untreated

$CATE(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$ conditional on covariates

- $ATE = \pi ATT + (1 - \pi) ATU$, where $\pi = \Pr(D_i = 1)$.
- They differ whenever effects are heterogeneous *and* related to who gets treated, which is the normal case in sociology.
- “Should we expand this program?” asks about the ATU. “Was it worth running?” asks about the ATT.
- Later we meet estimands nobody chose: the LATE (Week 7), the effect at a cutoff (Week 9), OLS’s own variance-weighted average (Week 6).

Estimands, With Numbers ★

Four units, effects that differ, and treatment related to the effect:

Unit	D_i	$Y_i(0)$	$Y_i(1)$	τ_i	Y_i observed
1	1	4	10	6	10
2	1	6	14	8	14
3	0	5	6	1	5
4	0	7	10	3	7

$$\text{ATE} = \frac{1}{4}(6 + 8 + 1 + 3) = 4.5, \quad \text{ATT} = \frac{1}{2}(6 + 8) = 7, \quad \text{ATU} = \frac{1}{2}(1 + 3) = 2.$$

- The naive comparison is $\frac{1}{2}(10 + 14) - \frac{1}{2}(5 + 7) = 6$, which equals none of them.
- $\text{ATT} + \text{selection bias} = 7 + (\frac{1}{2}(4 + 6) - \frac{1}{2}(5 + 7)) = 7 - 1 = 6$. ✓
- $\pi \text{ ATT} + (1 - \pi) \text{ ATU} = 0.5(7) + 0.5(2) = 4.5 = \text{ATE}$. ✓

Why the Naive Comparison Fails ★

The quantity we can compute is the difference in observed means. By the switching equation it is $E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]$; add and subtract $E[Y_i(0) \mid D_i = 1]$:

$$\underbrace{E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]}_{\text{naive comparison}} = \underbrace{E[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}$$

- Selection bias asks: *would the treated and untreated have differed even with no treatment?* Almost always yes.
- It involves **only** $Y_i(0)$, the world as it would have been for the treated. No amount of data on Y_i can settle it.

The Same Decomposition, Around the ATE

Write $\tau = \text{ATE}$ and $\pi = \Pr(D_i = 1)$. Since $\text{ATT} = \tau + (1 - \pi)(\text{ATT} - \text{ATU})$,

$$\begin{aligned}
 E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] &= \tau + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{selection on the baseline } Y_i(0)} \\
 &\quad + \underbrace{(1 - \pi)(\text{ATT} - \text{ATU})}_{\text{selection on the effect } \tau_i}.
 \end{aligned}$$

- Two ways to be wrong about the ATE: the treated would have done differently anyway, or the treated gain more from treatment. The four-unit table had both: baseline gap -1 , effect gap $7 - 2 = 5$, so $6 = 4.5 - 1 + 0.5 \times 5$.
- The second term is why “the treated are those who benefit most” is an identification problem, not a nuisance. Week 7’s LATE is what remains estimable when it is present.

In the Literature: Negative Selection (Brand and Xie 2010)

Treatment: completing a four-year degree. Outcome: log hourly wage. Data: NLSY79 ($n = 2,474$) and the Wisconsin Longitudinal Study ($n = 7,905$).

- Estimate each person's propensity to finish college from pre-college covariates (family background, measured ability, grades, aspirations); group people into balanced propensity strata; estimate the college effect within each stratum.
- The effect *falls* with the propensity. NLSY men at ages 29–32: about 30% higher wages in the lowest stratum, 10% in the highest, 5 points less per stratum. NLSY women at 37–40: 35% to 20%. Same sign in the Wisconsin cohort.
- In today's notation: τ_i is largest where $\Pr(D_i = 1 \mid X_i)$ is smallest, so $ATU > ATT$. The comparative-advantage story from economics predicts the opposite ordering.
- Consequence: the treated are the wrong people to learn the ATE from, and a policy that expands college reaches the ATU population.

Randomization Solves It ★

If D_i is assigned by a coin flip, it is independent of everything about the unit, including its potential outcomes:

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i.$$

Then $E[Y_i(0) \mid D_i = 1] = E[Y_i(0) \mid D_i = 0]$, the selection-bias term is zero, and

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] = \text{ATE} = \text{ATT}.$$

- Randomization does not make the groups identical; it makes them *exchangeable*, so any difference is chance. Experiments still adjust for pre-treatment covariates, for precision rather than bias.
- Randomizing an *offer* identifies the effect of the offer, not of the treatment taken up (Week 7).
- Every observational design tries to buy some version of this independence with an argument instead of a coin.

Estimating the ATE in an Experiment (Neyman 1923)

n_1 treated, n_0 controls, D_i by coin flip. Estimator $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$.

$$E[\bar{Y}_1] = E[Y_i(1) \mid D_i = 1] = E[Y_i(1)] \quad \text{consistency, then } D_i \perp\!\!\!\perp Y_i(1)$$

$$E[\hat{\tau}] = E[Y_i(1)] - E[Y_i(0)] = \text{ATE} \quad \text{same for } \bar{Y}_0$$

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \quad s_d^2 = \text{sample variance of } Y \text{ in arm } d$$

- Conservative: the true variance has a third term, $-\text{Var}(\tau_i)/n$, which no data can estimate because no unit shows both outcomes. Exact when effects are constant.
- It equals the HC2 robust variance from OLS of Y on D (Week 2), so `lm` plus a robust SE is Neyman's estimator (Samii and Aronow 2012).
- Simulated experiment, $n = 2,000$, $\text{ATE} = 4.03$: $\hat{\tau} = 4.08$, $\text{SE} = 0.116$. Over 2,000 re-randomizations the estimates have SD 0.102 and the SE averages 0.115: conservative, as promised.

SUTVA and Consistency: What the Notation Assumes ★

Writing “ $Y_i(1)$ ” at all assumes the **stable unit treatment value assumption**:

1. **No interference**: unit i 's outcome does not depend on anyone else's treatment. Otherwise each unit has 2^n potential outcomes.
2. **No hidden versions of treatment**: “ $D_i = 1$ ” means the same thing for everyone, or $Y_i(1)$ is ambiguous.

Consistency is the link from notation to data: for a unit that received d , the observed outcome *is* its potential outcome under d , $Y_i = Y_i(d)$ when $D_i = d$. It fails when the treatment has hidden versions, and under anticipation.

- Sociology is full of interference: peer effects, neighbourhood effects, diffusion. If everyone gets a degree, the return to a degree changes.
- Hidden versions: “college” covers a community-college certificate and a Princeton BA; “marriage” in 1960 and in 2020.
- State SUTVA and defend it, or redefine the treatment.

Beyond a Binary Treatment

Nothing above needs two arms. For a treatment d in a set $\mathcal{D}$, unit i has $Y_i(d)$ for every d and reveals $Y_i = Y_i(D_i)$.

- Years of schooling: $Y_i(12), Y_i(16), \dots$ “The effect of college” is $E[Y_i(16) - Y_i(12)]$, one contrast among many. A regression slope on years averages all the contrasts with weights nobody chose.
- Continuous dose: $d \mapsto E[Y_i(d)]$ is the dose-response curve. Positivity now asks for every dose inside every covariate cell.
- The estimands and the selection-bias decomposition go through one contrast at a time, so the course stays with $D_i \in \{0, 1\}$.

Identification Under Selection on Observables ★

Without randomization, assume it holds *within cells of observed covariates*.

The four conditions

1. **Conditional independence** (ignorability, unconfoundedness):
 $(Y_i(0), Y_i(1)) \perp\!\!\!\perp D_i \mid X_i.$
2. **Positivity** (overlap): $0 < \Pr(D_i = 1 \mid X_i = x) < 1$ wherever x occurs.
3. **Consistency**: $Y_i = Y_i(D_i).$
4. **SUTVA**: one unit's outcome does not depend on others' treatments, and "treated" means one thing for everyone.

Then the ATE is identified:

$$\text{ATE} = E \left[\underbrace{E[Y_i \mid D_i = 1, X_i] - E[Y_i \mid D_i = 0, X_i]}_{\text{estimable from data}} \right].$$

The left side is causal and unobservable; the right side is a function of the observed data. Conditions 1–4 are the bridge, and **none is testable**.

The Proof, Line by Line ★

Fix a covariate value x and work inside the cell $X_i = x$:

$$\begin{aligned}
 E[Y_i \mid D_i = 1, X_i = x] &= E[Y_i(1) \mid D_i = 1, X_i = x] && \text{consistency} \\
 &= E[Y_i(1) \mid X_i = x] && \text{conditional independence} \\
 E[Y_i \mid D_i = 0, X_i = x] &= E[Y_i(0) \mid X_i = x] && \text{same two steps} \\
 \text{CATE}(x) &= E[Y_i \mid D_i = 1, X_i = x] - E[Y_i \mid D_i = 0, X_i = x] && \text{subtract} \\
 \text{ATE} &= \sum_x \text{CATE}(x) \Pr(X_i = x) && \text{iterated expectations}
 \end{aligned}$$

- Positivity enters in line 1: $E[Y_i \mid D_i = 1, X_i = x]$ exists only if someone with $X_i = x$ is treated.
- Line 5 weights cells by how common they are; the ATT weights them by $\Pr(X_i = x \mid D_i = 1)$ instead.
- In missing-data language: the unobserved potential outcome is missing at random given X_i . Week 6 is line 4 when X has too many cells to take means.

What Positivity Is Saying ★

$0 < \Pr(D_i = 1 \mid X_i = x) < 1$: within every covariate cell we must see *both* treated and untreated units.

- If no one with $X = x$ is ever treated, no data in the world says what would happen if they were. Extrapolation there is model, not evidence.
- Positivity fails silently as covariates accumulate: with 10 binary controls there are 1,024 cells, and most are empty or one-sided. This is the curse of dimensionality for causal inference, and why Weeks 6 and 13 exist.
- Regression does not warn you. It fills empty cells by extrapolating the functional form.
- Diagnose it by looking at the estimated propensity score in the two groups (Week 6).

Path Analysis to DAGs

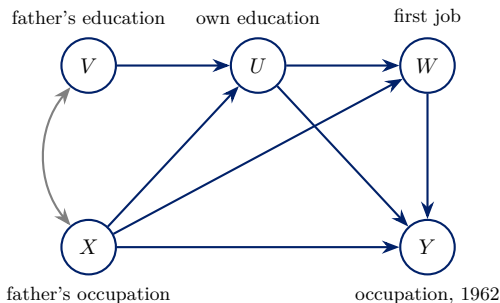
Causal Diagrams Are a Sociological Tradition

- The first formal treatment of causality in statistics was **Sewall Wright's** path analysis (Wright 1921).
- Sociology adopted it early. **Blau and Duncan** (1967) drew status attainment as a path diagram, father's education and occupation $\rightarrow$ respondent's education $\rightarrow$ first job $\rightarrow$ current occupation, and structural equation modelling grew out of it.
- Path analysis had **no definition of a causal effect** and no rule for which variables to include. The arrows were interpreted, not derived.
- **Pearl** (1995) supplied the missing half: a criterion, read off the graph, for *when* an effect is identified from observational data and what must be conditioned on.

DAGs are not an import into sociology. They are our own tradition, given its identification theory.

Theory Into a Graph: Blau and Duncan (1967)

The status-attainment model as they drew it: a theory of mechanisms as a picture, before any regression.



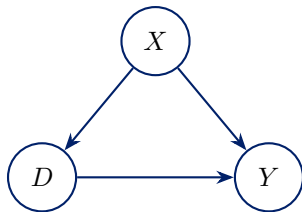
- Origins V , X are exogenous and correlated; the curved arrow makes no claim about why.
- Every arrow is a mechanism: origins reach the 1962 occupation through schooling and the first job, and directly.
- What the diagram cannot say: whether the effect of U on Y is identified. Something unmeasured (ability) may point into both.

What a DAG Is, and What It Leaves Out

Write down how each variable is generated. For a treatment D , an outcome Y , and a covariate X :

$$X := \epsilon_X, \quad D := f_D(X, \epsilon_D), \quad Y := f_Y(D, X, \epsilon_Y),$$

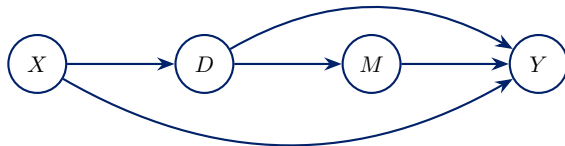
with independent ϵ 's. Draw $A \rightarrow B$ whenever A appears on the right of B 's equation.



- “ $:=$ ” means *is generated as*: a claim about what would happen if we intervened, not a correlation.
- **Acyclic**: no variable causes itself through a loop.
- The graph keeps which arrows exist and drops f_D, f_Y . Everything below holds for *any* functional form.

The Same Variable Can Play Two Roles

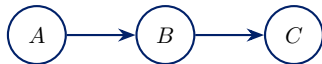
X causes D ; D causes M , which causes Y ; X and D also affect Y directly.



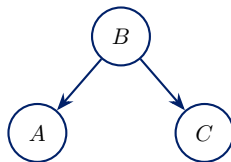
- Target: the effect of D on Y . Then X is a **confounder** (control for it) and M a **mediator** (do not).
- Target: the *total* effect of X on Y . Then D is a **mediator**, and controlling for it destroys the quantity you want.
- **Nothing in the data distinguishes the two cases.** The control set depends on the effect you want, so the estimand comes before the specification. No control set is correct for all questions at once.

DAGs

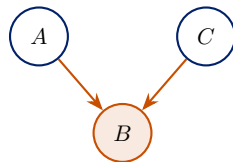
The Three Building Blocks ★



chain: B is a mediator



fork: B is a confounder



collider: B is a common effect

Structure	A and C , unconditionally	conditioning on B
Chain $A \rightarrow B \rightarrow C$	associated	blocks the path
Fork $A \leftarrow B \rightarrow C$	associated	blocks the path
Collider $A \rightarrow B \leftarrow C$	<i>independent</i>	opens the path

The collider row is the one that surprises people, and the source of most of the damage done by “controlling for everything.”

d -Separation ★

A *path* is any sequence of arrows joining two nodes, arrows pointing either way. Given a conditioning set Z , each path is open or blocked.

Blocking

A path is **blocked** by Z if it contains

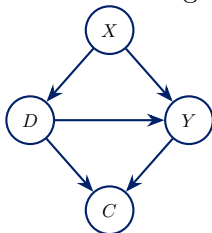
1. a chain $\cdot \rightarrow B \rightarrow \cdot$ or a fork $\cdot \leftarrow B \rightarrow \cdot$ with $B \in Z$, or
2. a collider $\cdot \rightarrow B \leftarrow \cdot$ with $B \notin Z$ and no descendant of B in Z .

A and C are **d -separated** by Z if every path between them is blocked.

- What it buys: d -separation by Z implies $A \perp\!\!\!\perp C \mid Z$ in every distribution the graph can generate. Within each stratum of Z , knowing A says nothing about C .
- This is a statement about the *observed* variables A, C, Z . No potential outcome appears; the connection to $Y(d) \perp\!\!\!\perp D \mid Z$ comes with the back-door criterion.
- Clause 2's tail: a descendant of a collider is a partial record of it. Conditioning on the descendant opens the path too.

d -Separation, Worked on One Graph

X : family background. D : degree. Y : earnings. C : employed at 40, caused by both degree and earnings.



Three paths from D to Y :

Path	$Z = \emptyset$	$Z = \{X\}$	$Z = \{X, C\}$
1: $D \rightarrow Y$	open	open	open
2: $D \leftarrow X \rightarrow Y$	open	blocked	blocked
3: $D \rightarrow C \leftarrow Y$	blocked	blocked	open

- Path 2, a fork at X : open until X is conditioned on. Path 3, a collider at C : blocked until C is conditioned on.
- D and Y are never d -separated: path 1 has no middle node. Right, since D causes Y .
- Identification does not ask for $D \perp\!\!\!\perp Y \mid Z$. It asks for paths 2 and 3 blocked with path 1 open: the back-door criterion, once do is defined.

Seeing Versus Doing ★

Two different distributions of Y , one symbol apart:

- $\Pr(Y \mid D = 1)$: Y among the units that *happened* to have $D = 1$. Seeing.
- $\Pr(Y \mid do(D = 1))$: Y if *every* unit were set to $D = 1$. Doing.

In the structural equations, $do(D = 1)$ replaces D 's equation and touches nothing else:

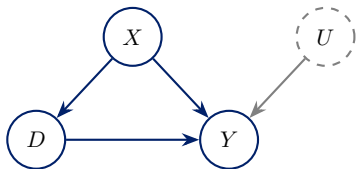
$$D := f_D(X, \epsilon_D) \longrightarrow D := 1 \quad \text{the intervention}$$

$$Y := f_Y(D, X, \epsilon_Y) \longrightarrow Y := f_Y(1, X, \epsilon_Y) = Y(1) \quad Y\text{'s equation unchanged}$$

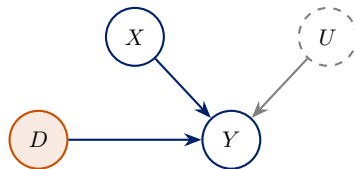
- The result is the potential outcome: $\Pr(Y \mid do(D = d)) = \Pr(Y(d))$, so $E[Y \mid do(D = 1)] - E[Y \mid do(D = 0)] = \text{ATE}$.
- Randomization: $D := \epsilon_D$, so the units with $D = 1$ are a random sample and $\Pr(Y \mid D = 1) = \Pr(Y(1))$. Seeing equals doing.
- Otherwise the units with $D = 1$ differ in X , and $\Pr(Y \mid D = 1) \neq \Pr(Y(1))$: the selection bias of the first section, in the new notation.

The Intervention Graph ★

Setting D deletes D 's equation, so it deletes every arrow *into* D . Arrows out of D stay.



how the data were generated



after $do(D = d)$

- The data come from the left graph. The effect is a contrast in the right graph, where D has no causes and so no back-door path.
- In the right graph D and X are d -separated: setting D does not change anyone's background, $\Pr(X \mid do(D = d)) = \Pr(X)$.
- Randomization makes the data come from the right graph. Otherwise, the next frame: find strata of X inside which the two graphs agree, and average.

From the Intervention Graph to a Formula ★

Previous graph, X observed. Goal: write $\Pr(Y \mid do(D = d))$ using only quantities from the left graph.

$$\begin{aligned}
 \Pr(Y \mid do(D=d)) &= \sum_x \Pr(Y \mid do(D=d), X=x) \Pr(X=x \mid do(D=d)) && \text{total probability} \\
 &= \sum_x \Pr(Y \mid do(D=d), X=x) \Pr(X=x) && D \text{ does not cause } X \\
 &= \sum_x \Pr(Y \mid D=d, X=x) \Pr(X=x) && \text{given } X, \text{ seeing} = \text{doing}
 \end{aligned}$$

- Line 3 needs the graph. Y 's equation is the same in both graphs; given X , the only difference between seeing $D = d$ and setting it is the back door $D \leftarrow X \rightarrow Y$, and X blocks it. Inside a stratum, a unit seen with $D = d$ is as good as one set to d .
- Means, then subtract: $E[Y \mid do(D=1)] - E[Y \mid do(D=0)] = \sum_x (E[Y \mid D=1, x] - E[Y \mid D=0, x]) \Pr(X=x)$, the formula of “The Proof, Line by Line,” now from the graph.
- U points only into Y , so it opens no path from D : harmless.

The Back-Door Criterion ★

Back-door criterion (Pearl 1995)

Z satisfies the back-door criterion for the effect of D on Y if

1. no node in Z is a descendant of D , and
 2. Z blocks every path between D and Y that starts with an arrow *into* D (a *back door*).
- In d -separation terms: delete every arrow *out of* D ; in what is left, Z must d -separate D from Y . The deletion removes causal and collider paths; only back doors remain. Condition 1 keeps mediators and colliders out of Z .
 - The previous derivation then holds with Z for X , the **adjustment formula**:
$$\Pr(Y \mid do(D=d)) = \sum_z \Pr(Y \mid D=d, Z=z) \Pr(Z=z).$$
 - Potential-outcome reading: the causes of $Y(d) = f_Y(d, \dots)$ are the arrows into Y other than D ; $Y(d) \perp\!\!\!\perp D \mid Z$ holds when Z blocks every path from D to them, condition 2. The criterion is the graphical test of conditional independence.

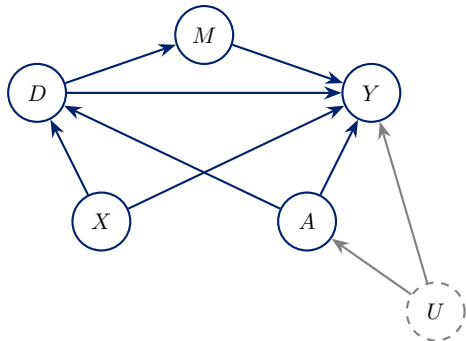
A Worked Example: College and Earnings

Does a college degree (D) raise earnings at 40 (Y)? Family background X and a measured test score A shape both; an unmeasured U (school quality) shapes the score and earnings; occupation M is one channel from degree to earnings.

Back-door paths from D to Y :

- $D \leftarrow X \rightarrow Y$: blocked by X .
- $D \leftarrow A \rightarrow Y$: blocked by A .
- $D \leftarrow A \leftarrow U \rightarrow Y$: blocked by A , a chain node on it.

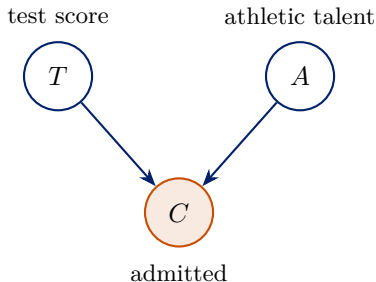
$\{X, A\}$ satisfies the criterion. U stays unmeasured and does no harm: every path through it runs through A . M is a descendant of D ; leave it out, or the estimate is at best the direct effect (next section).



Colliders

Colliders: The Intuition First ★

Admission to a selective college requires either a high test score or athletic talent; the two are independent among applicants.



- Among applicants, the test score says nothing about athletic talent: no arrow between T and A , no common cause.
- **Among admitted students**, someone with a mediocre score must have got in some other way, so they are probably a strong athlete. A *negative* association appears from nothing.
- This is Berkson's paradox, and the mechanism behind "selection bias" and the birth-weight paradox below.

The Algebra of a Collider ★

Give the picture equations, with independent $\mathcal{N}(0, 1)$ shocks:

$$T := \epsilon_T, \quad A := \epsilon_A, \quad C := T + A + \epsilon_C.$$

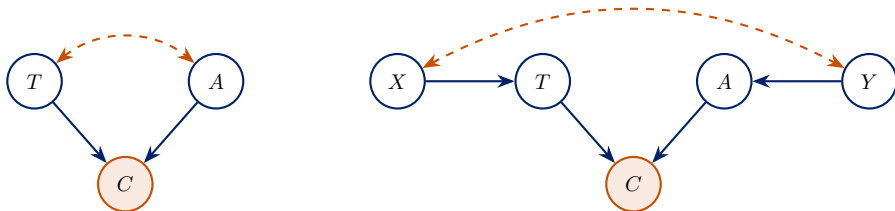
T and A are independent by construction: $E[T \mid A = a] = 0$ for every a . Now condition on C . Since $T = (C - A) - \epsilon_C$,

$$\begin{aligned} E[T \mid A, C] &= \frac{\text{Cov}(T, C - A)}{\text{Var}(C - A)} (C - A) && \text{projection of } T \text{ on } C - A = T + \epsilon_C \\ &= \frac{\text{Var}(T)}{\text{Var}(T) + \text{Var}(\epsilon_C)} (C - A) = \frac{1}{2} C - \frac{1}{2} A && \text{Cov}(T, T + \epsilon_C) = \text{Var}(T) = 1 \end{aligned}$$

- At fixed C , a unit more talent goes with **half a unit less test score**. The coefficient $-\frac{1}{2}$ is entirely spurious; nothing in the model connects A to T .
- Averaging over C given $A = a$ gives $E[T \mid A = a] = 0$ again, because $E[C \mid A = a] = a$. The bias lives only in the comparison at *fixed* C , which is what “controlling for C ” does.

The Damage Spreads to Ancestors

Conditioning on a collider opens associations among its parents **and all of their ancestors**.



- On the right, X and Y are four steps apart with no common cause. Conditioning on C makes them dependent anyway.
- A control can induce bias between variables it is not adjacent to. You cannot see this in a correlation table, only by tracing paths. Conditioning on a *descendant* of C does a milder version.

The Birth-Weight Paradox

A puzzle from US vital statistics (Hernández-Díaz, Schisterman, and Hernán 2006):

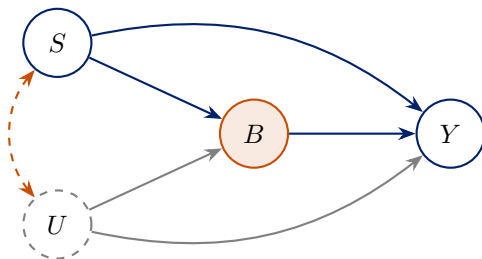
- Infants of smokers are more often of low birth weight (LBW).
- Infants of smokers have higher mortality.
- But among LBW infants, mortality is *lower* for infants of smokers than of non-smokers.

The naive reading: smoking protects low-birth-weight infants.

The structural reading: **low birth weight is a collider**. Smoking and unobserved competing risks (congenital abnormalities, severe maternal illness) both cause LBW, and those risks are far more lethal than smoking. Among LBW infants, “mother did not smoke” is evidence for “something worse.”

Same mechanism as admissions: two independent causes of the same event look negatively related once you condition on the event.

The Birth-Weight Paradox, Structurally ★



S smoking, B low birth weight, U competing risk, Y death

$$B := S + U + \epsilon_B, \quad Y := S + B + \kappa U + \epsilon_Y, \quad S := \epsilon_S, \quad U := \epsilon_U,$$

independent $\mathcal{N}(0, 1)$ shocks. Smoking's total effect on death is *harmful*: direct 1, plus 1×1 through birth weight, total +2. Regressing Y on S alone recovers +2. Conditioning on B opens $S \rightarrow B \leftarrow U$ (dashed).

Where the Sign Flips ★

Regressing Y on S and B , “controlling for birth weight”:

$$\begin{aligned}
 E[Y \mid S, B] &= S + B + \kappa E[U \mid S, B] && \epsilon_Y \text{ is independent of } S, B \\
 &= S + B + \kappa \cdot \frac{1}{2}(B - S) && \text{collider algebra: } U = (B - S) - \epsilon_B \\
 &= (1 - \frac{\kappa}{2})S + (1 + \frac{\kappa}{2})B.
 \end{aligned}$$

- The coefficient on S is no longer the direct effect 1. It has acquired $-\kappa/2$, entirely from the induced negative association between S and U within strata of B .
- If the competing risks are far more lethal than smoking, $\kappa \gg 2$, smoking looks **protective**.
- **Nothing is wrong with the estimator.** The regression reports a conditional expectation faithfully. The error is reading it as a causal effect.

Selection Bias Is Collider Bias ★

The most common invisible collider in sociology is *the sample itself* (Elwert and Winship 2014).

- **Wage regressions on employed workers:** employment is caused by schooling and by the unobserved determinants of wages. Restricting to the employed conditions on a collider.
- **Studies of college students, surviving firms, intact marriages:** each is a conditioning operation.
- **Panel attrition:** staying in the sample is an outcome.
- **“Among those who report X ”:** reporting is behaviour.

The practical rule

Draw the selection mechanism into your DAG as a node you have conditioned on. Then read off what it opens. If the sample was constructed by any process downstream of treatment or outcome, assume it opened something.

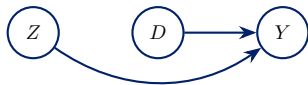
Controls

The Taxonomy ★

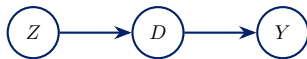
Variable	Role	Control for it?
Common cause of D and Y	confounder	yes ; this is the point
Pre-treatment predictor of Y only	precision variable	yes , reduces variance
Pre-treatment predictor of D only	instrument-like	<i>no</i> : inflates variance, amplifies bias
Consequence of D on the path to Y	mediator	no , removes part of the effect
Common effect of D and Y	collider	no , creates spurious association
Consequence of Y	outcome proxy	no
Sample selection on a collider	implicit conditioning	no , and often invisible

- Full taxonomy in Cinelli, Forney, and Pearl (2024).
- The third row again: a strong predictor of D with no direct effect on Y *amplifies* whatever unmeasured confounding remains. “More controls” is not conservative.

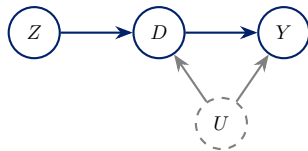
Causes of the Outcome Only, or of the Treatment Only



1: $Z \rightarrow Y$ only



2: $Z \rightarrow D$ only



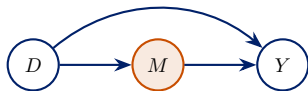
3: $Z \rightarrow D$ only, U unmeasured

- **1, good.** Z is a *precision* variable: it absorbs residual variance in Y and shrinks the standard error. Include it.
- **2, neutral for bias, bad for precision.** Z removes variation in D ; the estimate is unbiased but noisier.
- **3, bad.** Adjusting for Z throws away the clean variation in D and keeps the confounded part, so the remaining estimate leans harder on U : **bias amplification**. Variables like Z are *instruments* (Week 7); they identify an effect by a different route and must never be used as controls.

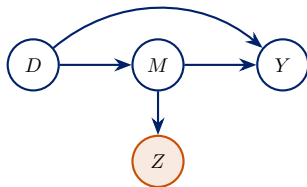
Post-Treatment Variables, and How They Sneak In

- Adjusting for a post-treatment variable is almost always a mistake, even when the coefficient “looks better.”
- The greater danger is implicit conditioning, through sample construction and variable definition rather than the regression formula:
 - the effect of education on wages, estimated on the *employed*;
 - occupational outcomes among people who *have* an occupation;
 - an outcome defined as a ratio whose denominator responds to treatment;
 - respondents who completed a follow-up wave.
- None of these appears in the model equation. All of them are conditioning.

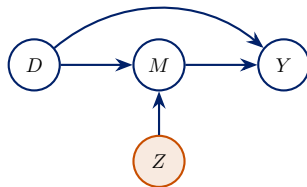
Mediators, Their Descendants, and Their Causes



control M : bad



control $Z \leftarrow M$: bad

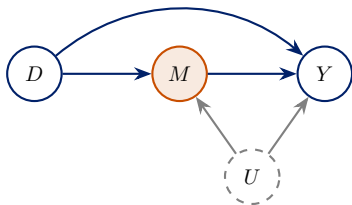


control $Z \rightarrow M$: neutral

- **Left.** M carries part of the effect of D . Holding it fixed removes that part; what remains is not the total effect.
- **Middle.** Z is a *descendant* of the mediator, a noisy copy of it. Holding Z fixed partly holds M fixed: the same damage, diluted.
- **Right.** Z causes the mediator but lies on no path from D to Y . Controlling for it leaves the total effect alone. The rule is about *paths*, not timing alone.

Two Sharper Cases

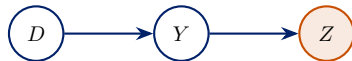
Mediators and the direct effect



M is a bad control *even if you want the direct effect*: conditioning on M opens $D \rightarrow M \leftarrow U \rightarrow Y$. Direct effects need *stronger* assumptions than total effects.

No statistical test distinguishes any of these from a confounder. The distinction is causal, and it must come from the DAG.

Consequences of the outcome



Z happens *after* Y and is caused by it. Controlling for it partials out variation in the outcome itself. The direction of the bias depends on the rest of the graph; do not assume it is conservative.

The Mechanism Test: Baron and Kenny (1986) ★

The standard way sociology “tests a mechanism”: show that the effect of D on Y shrinks once the mediator M is controlled. Three regressions:

$$Y = cD + e_1$$

total effect c

$$M = aD + e_2$$

D moves the mediator

$$Y = c'D + bM + e_3$$

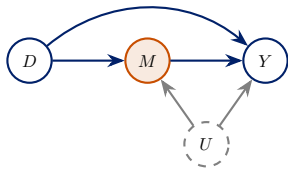
direct effect c' , mediator effect b

- With OLS and no interaction, $c = c' + ab$ holds *exactly*: Week 1’s omitted-variable formula with M omitted. ab is the **indirect effect**, ab/c the “share mediated.” Sobel’s SE for ab , $\sqrt{b^2\text{se}(a)^2 + a^2\text{se}(b)^2}$, is Week 3’s delta method.
- GSS, parents’ education $\rightarrow$ own education $\rightarrow$ log earnings (sex, race, age held): $c = 0.063$, $a = 0.355$, $b = 0.111$, $c' = 0.023$, so $ab = 0.039$: “63 percent mediated.”
- Simulation with $a = 0.8$, $b = 1.5$, $c' = 1$ returns 2.2, 0.8, 1.5, 1.0. The method works when its assumption holds.

What the Mechanism Test Assumes ★

Step 3 regresses Y on M : a causal claim about $M \rightarrow Y$ that needs its own back door closed, **no unmeasured cause of both M and Y** . Randomizing D does not deliver it.

Same simulation with $U \rightarrow M$, $U \rightarrow Y$ added, U unmeasured:



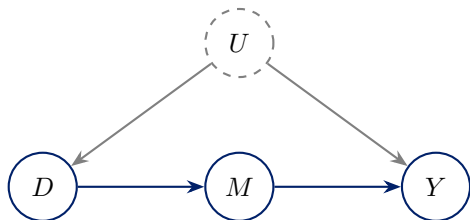
	c	a	b	c'	ab	share
truth	2.2	0.8	1.5	1.0	1.2	0.55
Baron–Kenny	2.19	0.79	2.00	0.60	1.58	0.72

- c and a survive: D has no back door. b absorbs U ; the indirect effect is overstated, the direct understated. The sum is still c ; the split is wrong.
- GSS: ability points into own education and into earnings; “63 percent mediated” rests on an assumption the data cannot check.
- Modern name: *sequential ignorability* (Imai, Keele, and Tingley 2010). State it; add a sensitivity analysis (`mediation::medsens`: GSS indirect effect is zero once U explains 9% of the variance of both M and Y).

Front Door

Identification Without Measuring the Confounder

Sometimes no back door can be closed, but the *mechanism* can be measured.



The three conditions

1. **Complete mediation:** every directed path from D to Y passes through M .
2. **No open back door from D to M .**
3. **Every back door from M to Y is blocked by D .**

U is unobserved and confounds D and Y , yet the effect of D on Y is identified. A surprising result, worth understanding rather than memorizing.

Why It Works: Two Identified Legs

Split $D \rightarrow Y$ into two effects, each of which *is* identified.

- **Leg 1**, $D \rightarrow M$. By condition 2 there is no back door, so $M(d)$ is independent of D and $\Pr(M(d) = m) = \Pr(M = m \mid D = d)$. Read it straight off the data.
- **Leg 2**, $M \rightarrow Y$. The only back door from M to Y is $M \leftarrow D \leftarrow U \rightarrow Y$, and D is a chain node on it. Conditioning on D blocks it, so the effect of M on Y is identified by adjusting for D .
- **Compose**. The effect of D on Y is the effect of D on M , pushed through the effect of M on Y ; by complete mediation there is nothing else.

Neither leg requires observing U . The confounder sits outside both sub-problems, and that is the whole trick.

The Front-Door Formula ★

$$\begin{aligned}
 \Pr(Y \mid do(D = d)) &= \sum_m \Pr(M = m \mid do(D = d)) \Pr(Y \mid do(M = m)) \\
 &= \sum_m \Pr(M = m \mid D = d) \Pr(Y \mid do(M = m)) \\
 &= \sum_m \Pr(M = m \mid D = d) \sum_{d'} \Pr(Y \mid M = m, D = d') \Pr(D = d')
 \end{aligned}$$

- Line 1: complete mediation, D reaches Y only through M . Line 2: leg 1, no back door from D into M . Line 3: leg 2, the back-door adjustment for $M \rightarrow Y$ with $Z = D$.
- The inner sum averages over the *marginal* distribution of D , note d' and not d . In potential outcomes the left side is the distribution of $Y(d)$.
- Everything on the right is a proportion in the data; U never appears. In linear models: the slope of M on D times the slope of Y on M controlling for D .

A Sociological Application: the Age–Period–Cohort Problem

Attitudes change across time. Is that because people *age*, because the *period* changed everyone, or because *cohorts* raised differently replace one another? The standard model,

$$Y_{it} = \alpha + \beta_A A_{it} + \beta_P P_t + \beta_C C_i + \epsilon_{it}, \quad C_i = P_t - A_{it},$$

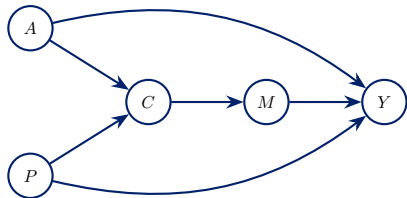
cannot answer. Because cohort is period minus age, the three regressors are perfectly collinear, and for any δ

$$(\beta_A + \delta)A + (\beta_P - \delta)P + (\beta_C + \delta)(P - A) = \beta_A A + \beta_P P + \beta_C (P - A)$$

fits exactly as well. A one-parameter family of “effects” explains the data identically. Decades of proposed fixes, equality constraints, the intrinsic estimator, choose a point on that family by assumption. They are conventions, not identification.

Mechanism-Based Identification for APC

Winship and Harding (2008) apply the front door. Suppose the *cohort* effect works entirely through measured mechanisms M : cohort size, educational attainment, the labour market at entry.



- **Leg 2**, $M \rightarrow Y$ given age and period: the outcome equation has A , P , M and no C . No collinearity.
- **Leg 1**, $C \rightarrow M$: identified when the mechanism equation *excludes* age or period on substantive grounds. Completed schooling is fixed by 30 and ignores the current period.
- **Compose**: the cohort effect is leg 1 pushed through leg 2.

The collinearity was never the binding constraint. A stated mechanism, with an exclusion per leg, identifies a quantity the linear model could not even define.

Critiques of Front-Door Identification

Be sceptical of front-door claims, including your own.

- **Complete mediation is a strong claim.** Every effect of D on Y must pass through the *measured* M . One unmediated path breaks identification; a cohort plausibly acts through many unmeasured channels at once.
- **No hidden confounding of the legs.** $D \rightarrow M$ must have no back door, and $M \rightarrow Y$ must be identified by D alone. Anything unobserved that drives both the mechanism and the outcome breaks leg 2.
- **Measurement.** In leg 1, M is the outcome and classical error only adds noise. In leg 2, M is conditioned on, and error there biases the leg and the composed estimate.
- **Sensitivity.** The assumptions are strong: report how the estimate moves as a small unmediated path is allowed.

A real result, rarely satisfied. Its value is clarifying *what you would need to know* to make a mechanism argument work.

Potential Outcomes and DAGs Are Complements

Potential outcomes

- Precise about the *estimand*: ATE, ATT, LATE.
- Natural home for heterogeneity, effect distributions, and design-based inference.
- Clumsy for reasoning about long chains of variables.

DAGs

- Precise about *identification*: which paths are open, what to condition on.
- Makes collider and mediator mistakes visible immediately.
- Silent about magnitude, heterogeneity, and which average you get.

Use both

Draw the DAG to work out *what to condition on*. Write the potential outcomes to state *what you are estimating*. They are one theory in two notations; in practice each makes a different mistake easy to catch.

Estimands

The Standard Workflow Skips a Step ★

Lundberg, Johnson, and Stewart (2021): most quantitative papers jump from theory to a regression, leaving the target quantity implicit. Their sequence:

1. **Theoretical estimand.** A *unit-specific quantity* and a *target population*. “The average difference in earnings at 40 between completing college and not, among US-born adults who completed high school.” Stated before any data.
2. **Empirical estimand.** Which identification assumptions turn the theoretical estimand into a function of observable distributions? This is the back-door argument.
3. **Estimation strategy.** Only now: regression, matching, weighting, IV, machine learning.

Most methodological disputes are disagreements at step 1 or 2 conducted as if they were about step 3.

Why This Discipline Pays

- It shows when the estimand was chosen *by the method*: OLS's variance-weighted average (Week 1), the LATE for compliers (Week 7), the effect at the cutoff (Week 9).
- It makes “the coefficient on X_3 in Model 4,” a quantity nobody can state in words, an unacceptable answer.
- It separates two objections to a paper: “your estimand is uninteresting” and “your estimate does not recover your estimand.”
- The target population is part of the estimand, so extrapolation is a stated claim rather than an implicit one.

A habit worth forming

Write the theoretical estimand as one sentence, in words, before you write any code. You will be asked to do exactly this on the final exam.

Where This Leaves Us

- Causal inference is a missing data problem. We target averages because individual effects are unknowable.
- The naive comparison equals the ATT plus selection bias. Every design in this course is an argument *about* that term: adjustment drives it to zero (Week 6), or another comparison sidesteps it at the price of a narrower estimand (Weeks 7–11).
- Conditional ignorability, positivity, consistency, and SUTVA identify the ATE. **None is testable.**
- A DAG turns those assumptions into a picture you can check for colliders and mediators.

Coming next

Week 6: if we believe conditional ignorability, how should we adjust? Matching, propensity scores, weighting, doubly robust estimation, and why regression is one option among several. **Weeks 7–11:** what to do when we do *not* believe it.

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